# Joe Wlos

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## PROFESSIONAL EXPERIENCE

**INTRVL:** *Manager, Product and Analytics Engineering* **June 2021 – Present**

- Led engineering, creative, and data pipeline development for ad surveys deployed on programmatic advertising networks, text-2-web, and Lucid to measure the effectiveness of liberal groups' online media
  - Captured survey responses from over 90,000 different individuals across a dozen client projects
  - Presented modeled results–MRPs and casual inference analyses–in conjunction with data science team
- Visualized COVID-19 vaccine surveys on both internal and public dashboards, developed in React
  - Built dynamic shareable infographics to display vaccine hesitancy rates at state- and county-levels
  - Finalist for the 2022 Reed Award for "Best Data Visualization"

**Hawkfish:** *Data Scientist, Voter Research and Business Intelligence* **February 2020 – May 2021**

- Identified and communicated "Red Mirage" concept to counter disinformation about vote-by-mail
  - Built online scenario tool–combining state polling, survey data on voting method by candidate choice, and research on state election procedures–to estimate horserace reporting on election night and beyond
  - Conveyed findings to Democratic Party decision makers, media organizations, and the public, resulting in over 2,000 mainstream press reports and 200,000 tweets
- Developed web application for creative teams and marketers to assess second-to-second viewer attention and emotion in pre-testing of video ads, enabling better ad targeting and content analysis

**Cello Capital Management:** *Consultant* **December 2019 – January 2020**

- Refactored Python and SQL for commercial mortgage-backed securities (CMBS) analytics in Power BI

**Dwight Securities Management:** *Associate* **June 2017 – October 2019**

- Traded and analyzed CMBS issued by federal agencies and enterprises (Ginnie Mae and Fannie Mae)
  - Built property valuation and cashflow models in Python and VBA to support traders and assess risk
  - Created integrated dashboards for visualizing and searching multifamily and nursing facility datasets

**JOOR:** *Data Analyst* **June 2016 – May 2017**

- Developed histogram of oriented gradients feature descriptors to classify clothing categories in images
- Used k-means clustering of pixels in product photos to produce visualizations of dominant seasonal colors

**Morningstar:** *Data Analyst, Reference Data* **July 2015 – May 2016**

- Wrote Python scripts to map identifiers to derivatives using Bloomberg's Open FIGI API

## EDUCATION

**Grinnell College:** *B.A. in Political Science with Honors* **Class of 2015**

- Thesis applying historical development of *habeas corpus* to extrajudicial detentions at Guantánamo Bay
- Winner of 2015 Baumann Essay Prize for research examining the effective regulation of net neutrality

## TECHNICAL SKILLS

**Data Engineering**
- Airflow
- DBT
- SQL

**Data Analysis**
- Pandas
- NumPy & SciPy
- VBA

**Dashboards**
- Google Data Studio
- Plotly & Dash
- Power BI

**Web Development**
- Flask & Django
- React
- Tailwind CSS